IR Evaluation

- why not accuracy?
  → class imbalance issue.
  → accuracy ignores ranking
  → gives equal wts. to FP & FN
    ↳ by in IR → missing relevant info. is worse than showing few irrelevant things.
  → doesn't reflect user experience

- it is not good to judge a system with precision & recall values. As they might use diff. threshold settings. It will be more informative to compare ranking vs ranking.
- search engine produces a set & not ranking. Later, we can calculate precision & recall for every rank.

- MAP (Mean Avg. Precision)

1. for single query

$$\boxed{AP = \frac{1}{|Rel|} \sum_{k=1}^{n} P(k) \cdot rel(k)}$$

precision at rank k

total relevant docs

↳ 1 if doc. at k is relevant

2. $$\boxed{MAP = \frac{1}{Q} \sum_{q=1}^{Q} AP(q)}$$

→ measures how well a sys. orders relevant docs across multiple queries.

- nDCG (normalized discounted cumulative gain)
  ↳ measures how good your ordering is, when relevance is graded.

1. $$\boxed{DCG = \sum_{i=1}^{n} \frac{2^{rel_i} - 1}{\log_2 (i+1)}}$$

→ gives more wt to relevant docs

→ discounts lower rank

rel_i = relevance score at pos i

i = rank pos.

2. $$nDCG = \frac{DCG}{iDCG}$$

↳ ideal rank

- MRR (mean reciprocal rank)
  ↳ how quickly the first relevant result appears.

$$\boxed{MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i}}$$

Q → no. of queries

→ rank of 1st relevant doc.